

Chapter Two: Diesel Engine Condition Monitoring

2.1 Introduction

Condition monitoring of diesel engines can be assessed on a continuous or periodic basis from observation or measurement of selected parameters. The application of CM and fault diagnosis strategies to a diesel engine is a well-recognized method of increasing its operational efficiency and reducing consequential damage, spare parts inventories and breakdown maintenance.

The main job of most monitoring systems is to obtain information about the engine in the form of primary data and, through the use of modern signal processing techniques, to provide vital information to the engine operator and the engine control system, before any failure occurs with the engine in service. Good monitoring systems for diesel engines can achieve at least the following benefits:

- Improved decision-making capability for selection of optimum engine operation conditions.
- Only defective equipment or assemblies are replaced, reducing time and cost of maintenance.
- Effective prediction and planning of maintenance operations. Time scheduled for maintenance can be used effectively since the nature of the fault is known in advance and both spare parts and labor can be organized accordingly.
- Reduction of engine emissions and fuel consumption.

2.2 Principal Faults in Diesel Engines

The diesel engine when operating normally can give thousands of hours of uninterrupted service. However, if a fault develops, the growth of the fault tends to be fairly rapid and can lead to major failure which can cause loss of life, damage to property and incur high costs when it occurs in, for example, commercial transport vehicles or ships. This why, it is essential to implement reliable and sensitive engine condition monitoring techniques.

Diesel engines use high compression ratios, generating a sufficiently high pressure and temperature to cause spontaneous ignition of the injected fuel. Also, the speed of engine rotation is 3000 rpm or more. The high speed, high pressure and high temperature increase the risk of faults occurring within the engine.

One of the most important elements is fuel injection system malfunction, which is responsible for about 43% of the engine faults. These system faults can also directly affect engine efficiency, exhaust emission, engine noise and other parameters.

Another important element is the lubrication system malfunction account for a high percentage of the engine faults. The lubrication system faults can directly affect the engine power, emission and other performance parameters, this is why it worthwhile to study this element of the system and associated faults.

In this work only faults in the combustion system are studied

Injection Timing Early (ITE)

Injection Timing Late (ITL)

Injection Valve nozzle wear

Piston Ring Wear (PRW)

Piston Ring Blow-by

Exhaust Valve Leaking

Liner Crack

Fuel Oil pump

Fuel Oil Pump Wear

Scavenge airport deposits (SAPD)

Scavenge air box fire

FO High Pressure Pipe rupture

Only the following faults are investigated:

- ITE
- ITL
- PRW
- SAPD

Healthy readings are taken before and after each fault

All faults are introduced with increasing severity from 0 to 90%

2.2.1 Injection Time Early:

Will cause early firing in a cylinder and a very high peak pressure at about top center of the piston stroke. This will cause a heavy shock load to be transmitted to the bearing with a corresponding knock from the engine.

2.2.2 Injection Timing Late:

Will cause late combustion of fuel will take place during the expansion stroke of the engine cycle. It causes losses in power, since the fuel is not burned correctly to transmit energy to the piston

2.2.3 Piston Ring Wear:

Will cause high rate of cylinder liner wear due to poor lubrication, and cause scavenge fire. There is also a risk of sized piston due to local overheating. There will be low compression and consequently poor combustion.

2.2.4 Scavenge air-port deposits:

Caused by unburned oil and carbon, this may include unburned fuel or lubricating oil and may be due to incorrect combustion caused by a defective injector, faulty fuel pump timing, incorrect fuel condition, low compression.

2.3 Experimental readings

After reaching the steady state, in either the healthy and the faulty case, the engine performance parameters are printed, and the following sets of readings are taken:

- Fuel Index
- Engine Speed (RPM)
- Mean Indicated Pressure (MIP), bar
- Indicated Power (IKW), KW
- Compression Pressure (PCOM)
- Max Pressure (PMAX)
- Time of start of injection (TING)
- Time of max pressure (TMAX)
- Exhaust outlet temperature (EXH)
- Exhaust outlet temperature deviation

Chapter Three: Data Processing and Fault Analysis

3.1 Introduction

The objective of this study is to develop a fault identification system based on the investigations made on the marine diesel engine and the faults to be identified as explained in chapter 2.

The data follow is obtained from the experiments carried out from previous work.

From these data, faults identifiers are to be selected from the engine performance parameters, that are most affected by the faults.

- The effect of the fault on the identifier is substantial, so that minimum number of identifiers would be sufficient to identify the fault.
- The identifier can be easily measured with simple instruments available on board ship

3.2 Data recorded

After setting the fault according to the marine engine studies in the last chapter, and obtaining the data of the deviation of each variable taken for each case of the engine performance.

The sets of readings extracted and obtained from the previous work are tabulated bellow.

MIP DIV%	IKW DIV%	PCOM DIV%	PMX DIV%	EXH DIV%		Severity	Fault Type
-0.2	0.2	0.3	0.2	-0.5		0	HE
1.6	1.7	0	0.9	-1.5		10%	ITE
2.7	2.8	0.05	1.5	-2.7		20%	ITE
4.5	4.5	0	2.5	-4		30%	ITE
5.8	5.8	0	3.2	-5.2		40%	ITE
7.5	7.5	0	4.3	-6.3		50%	ITE
8.8	8.8	0	5	-7.5		60%	ITE
10.2	10.2	0	5.6	-9.2		70%	ITE
11.6	11.6	0	6	-10.1		80%	ITE
13	13	0	6.5	-11.3		90%	ITE
-0.4	-0.6	-0.7	-0.5	0		0	HE
-2.2	-2.2	0	-0.8	1.2		10%	ITL
-3.9	-3.9	0	-1.5	2.3		20%	ITL
-6.2	-6.2	-0.3	-2.8	4.8		30%	ITL
-8	-8	0	-3.8	5.2		40%	ITL
-10.5	-10.5	-0.2	-5.2	6.8		50%	ITL
-12.5	-12.5	-0.4	-6.5	8.5		60%	ITL
-14.7	-14.7	-0.4	-8	9.5		70%	ITL
-17	-17	-0.6	-9.6	10.4		80%	ITL
-19.2	-19.2	-0.4	-11	11.3		90%	ITL
0.4	0	0.4	0.6	-0.4		0	HE
-1.6	-1.5	-1.1	-1	0.2		10%	PRW
-2.7	-2.5	-2.4	-2.2	1.4		20%	PRW
-2.2	-2.8	-2.3	-2.1	2.8		30%	PRW
-2.5	-4	-3.2	-2.2	4		40%	PRW
-5.1	-5.2	-4.2	-5.1	5.2		50%	PRW
-6.2	-6.8	-5.8	-5.9	6.1		60%	PRW
-7.2	-8.2	-7.3	-7.1	7.7		70%	PRW
-6.7	-8.7	-7.1	-6.4	8.8		80%	PRW
-9.2	-10.8	-9.6	-9.2	10.5		90%	PRW
-0.5	0	0	-0.1	0.5		0	HE
-2	-2	-1	-1	4		10%	SAPD
-3	-3	-2.5	-2	8.5		20%	SAPD
-4.5	-4.5	-4	-3	13.5		30%	SAPD
-6.5	-6.5	-5.5	-4.5	19.5		40%	SAPD
-8	-8	-6.5	-5	25.5		50%	SAPD
-8.5	-8.5	-7.5	-6.5	33		60%	SAPD
-10.5	-10.5	-8.5	-7.5	40.5		70%	SAPD
-12	-12	-10	-8.5	49		80%	SAPD
-14	-14	-12	-9.5	59.5		90%	SAPD

Table 3.1 : the percentage of the deviation of each variable and its fault type and severity

3.3 Fault identifiers

From the table of results to study the effect of fault types and their severity on the marine engine parameters, it became apparent that these values can be taken as identifiers, as they are the most affected by the faults in this study.

These identifiers are the percentage deviation of:

- Mean Indicated Pressure (MIP), bar
- Indicated Power (IKW), KW
- Compression Pressure (PCOM), bar
- Max Pressure (PMAX), KW
- Exhaust outlet temperature (EXH), °C

PCOM, PMX and EXH are easy to measure with simple instruments available on board ship, while MIP and IKW require special instrument.

3.3.1 Analysis

- Percentage deviation of PCOM is neither affected by ITE nor by ITL.
- The effect of ITE on the percentage deviation of MIP, IKW, PMX and EXH is the opposite of the effect of ITL
- Percentage deviation of EXH increases in case of ITL, PRW and SAPD.
- Percentage deviation of PCOM decreases in case of PRW and SAPD.

3.3.2 Process

It is noticed from the analysis that the effects of ITL, PRW and SAPD are somewhat similar with respect to their identifiers, but with different rates.

Further investigation has to be made to create an identifier using an Artificial Neural Network Model (ANN). These are discussed in chapter 5.